

The Law of Sines (Part II): The Ambiguous Case (SSA)

Solving a Triangle: Finding all three side lengths and all three interior angle measures of a triangle.

Scenarios when Solving Triangles

The Law of Cosines

SSS SAS

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = c^2 + a^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

The Law of Sines

AAS ASA SSA

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Ambiguity

i) No triangle

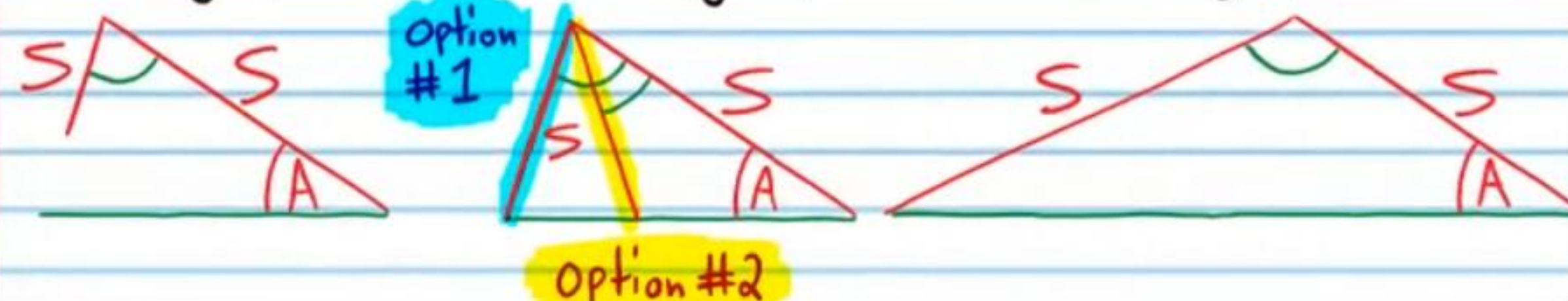
ii) One triangle

iii) Two triangles

The Side-Side-Angle Scenario

Red = Does not change Green = Can change

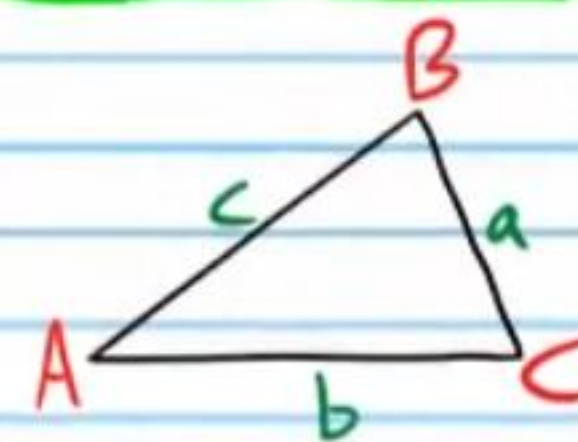
No triangle possible Two triangles possible One triangle possible



The Law of Sines

If a , b , and c are the side lengths of a triangle, and A , B , and C are the interior angle measures, then the equation to the right is true.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

- ① $C = 3\text{mm}$, $a = 1.5\text{mm}$, $A = 44^\circ$

I) Find $m\angle C$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 44^\circ}{1.5} = \frac{\sin C}{3}$$

$$(3) \frac{\sin 44^\circ}{1.5} = \frac{\sin C}{3} (3)$$

$$\frac{3 \sin 44^\circ}{1.5} = \sin C$$

$$\sin C \neq 1.389$$

No Triangle Possible

② $b = 8 \text{ ft}$, $a = 10 \text{ ft}$, $B = 130^\circ$

I) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{10} = \frac{\sin 130^\circ}{8}$$

$$10 \cdot \frac{\sin A}{10} = \frac{\sin 130^\circ}{8} \cdot 10$$

$$\sin A = \frac{10 \sin 130^\circ}{8}$$

$$\sin A \approx 0.9575$$

$$A \approx \sin^{-1}(0.9575...)$$

$$A \approx 73.2468^\circ$$

$$130^\circ + 73.2468^\circ \nless 180^\circ$$

$$203.2^\circ \nless 180^\circ \quad \text{X}$$

No Triangle Possible

③ $b = 27 \text{ in}$, $c = 63 \text{ in}$, $C = 120^\circ$

I) Find $m\angle B$.

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin B}{27} = \frac{\sin 120^\circ}{63}$$

$$\sin B = \frac{27 \sin 120^\circ}{63}$$

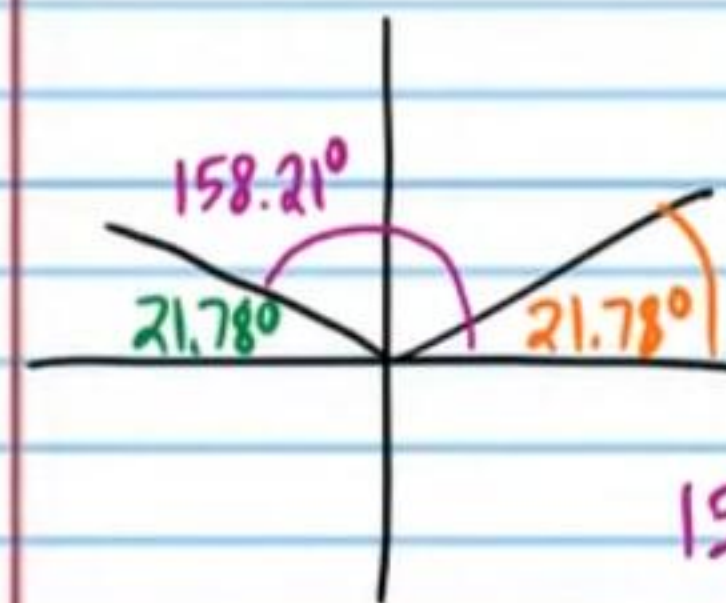
$$\sin B \approx 0.3711537...$$

$$B \approx \sin^{-1}(0.3711537...)$$

$$B \approx 21.7867^\circ$$

$$21.7867^\circ + 120^\circ < 180^\circ$$

$$141.78^\circ < 180^\circ \quad \checkmark$$



$$158.21^\circ + 120^\circ \nless 180^\circ$$

$$278.21^\circ \nless 180^\circ \quad \text{X}$$

One Triangle Possible

II) Find $m\angle A$.

$$A + B + C = 180^\circ$$

$$A + 21.7867^\circ + 120^\circ \approx 180^\circ$$

$$A \approx 38.2133^\circ$$

III) Find a .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 38.2133^\circ}{a} \approx \frac{\sin 120^\circ}{63}$$

$$a \approx \frac{63 \sin 38.2133^\circ}{\sin 120^\circ}$$

$$a \approx 45 \text{ in}$$

- ④ $b = 8 \text{ km}$, $a = 12 \text{ km}$, $B = 40^\circ$

I) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{12} = \frac{\sin 40^\circ}{8}$$

$$\sin A = \frac{12 \sin 40^\circ}{8}$$

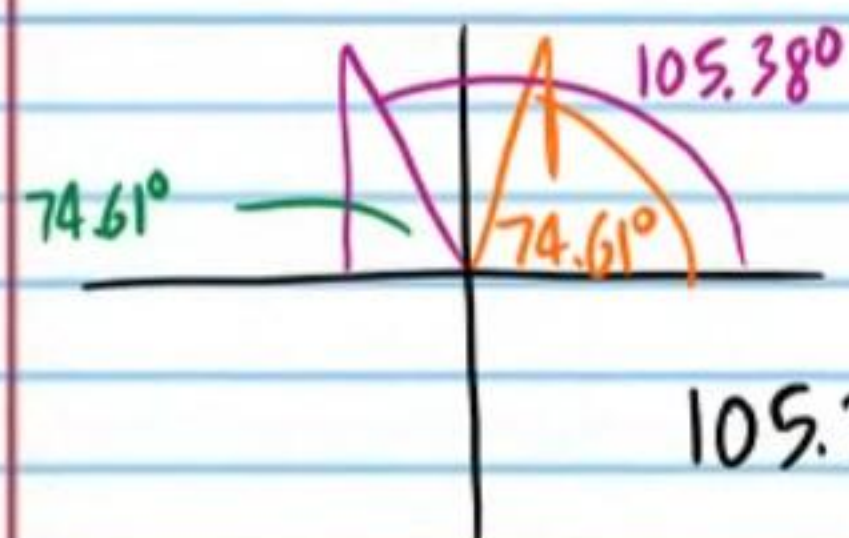
$$\sin A \approx 0.96418...$$

$$A \approx \sin^{-1}(0.96418...)$$

$$A \approx 74.6185^\circ$$

$$74.6185^\circ + 40^\circ < 180^\circ$$

$$114.618^\circ < 180^\circ \checkmark$$



$$A \approx 105.3814^\circ$$

$$105.38^\circ + 40^\circ < 180^\circ$$

$$145.38^\circ < 180^\circ \checkmark$$

Two Triangles Possible

II) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$74.6185^\circ + 40^\circ + C \approx 180^\circ$$

$$C \approx 65.3815^\circ$$

III) Find c .

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 65.3815^\circ}{c} \approx \frac{\sin 40^\circ}{8}$$

$$c \approx \frac{8 \sin 65.3815^\circ}{\sin 40^\circ}$$

$$c \approx 11.31 \text{ km}$$

$$b = 8 \text{ km}, a = 12 \text{ km}, B = 40^\circ$$

$$A \approx 74.6185^\circ, C \approx 65.3815^\circ$$

$$c \approx 11.31 \text{ km}$$

II) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$105.3814^\circ + 40^\circ + C \approx 180^\circ$$

$$C \approx 34.6186^\circ$$

III) Find c .

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 34.6186^\circ}{c} \approx \frac{\sin 40^\circ}{8}$$

$$c \approx \frac{8 \sin 34.6186^\circ}{\sin 40^\circ}$$

$$c \approx 7.07 \text{ km}$$

$$b = 8 \text{ km}, a = 12 \text{ km}, B = 40^\circ$$

$$A \approx 105.3814^\circ, C \approx 34.6186^\circ$$

$$c \approx 7.07 \text{ km}$$

⑤ $b = 4 \text{ ft}$, $a = 9 \text{ ft}$, $B = 59^\circ$

I) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{9} = \frac{\sin 59^\circ}{4}$$

$$\sin A = \frac{9 \sin 59^\circ}{4}$$

$$\sin A \approx 1.9286$$

No Triangle Possible

⑥ $A = 110^\circ$, $c = 180 \text{ mm}$, $a = 170 \text{ mm}$

I) Find $m\angle C$.

$$\frac{\sin C}{c} = \frac{\sin A}{a}$$

$$\frac{\sin C}{180} = \frac{\sin 110^\circ}{170}$$

$$\sin C = \frac{180 \sin 110^\circ}{170}$$

$$\sin C \approx 0.9949...$$

$$C \approx \sin^{-1}(0.9949...)$$

$$C \approx 84.25^\circ$$

$$84.25^\circ + 110^\circ \nless 180^\circ$$

$$194.25^\circ \nless 180^\circ \quad \times$$

No Triangle Possible

⑦ $C = 38^\circ$, $c = 10\text{cm}$, $a = 7\text{cm}$

I) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{7} = \frac{\sin 38^\circ}{10}$$

$$\sin A = \frac{7 \sin 38^\circ}{10}$$

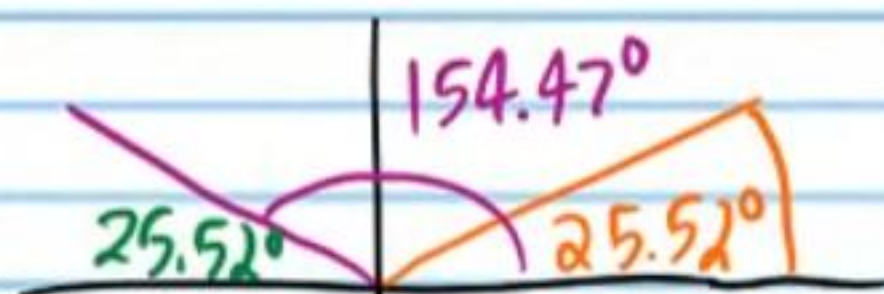
$$\sin A \approx 0.43096...$$

$$A \approx \sin^{-1}(0.43096...)$$

$$A \approx 25.52869^\circ$$

$$25.52^\circ + 38^\circ < 180^\circ$$

$$63.52^\circ < 180^\circ \checkmark$$



$$154.47^\circ + 38^\circ \neq 180^\circ$$

$$192.47^\circ \neq 180^\circ \times$$

II) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$25.52869^\circ + B + 38^\circ \approx 180^\circ$$

$$B \approx 116.4713^\circ$$

III) Find b .

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 116.4713^\circ}{b} \approx \frac{\sin 38^\circ}{10}$$

$$b \approx \frac{10 \sin 116.4713^\circ}{\sin 38^\circ}$$

$$b \approx 14.54\text{cm}$$

⑧ $b=20\text{m}, c=11\text{m}, C=20^\circ$

I) Find $m\angle B$.

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin B}{20} = \frac{\sin 20^\circ}{11}$$

$$\sin B = \frac{20 \sin 20^\circ}{11}$$

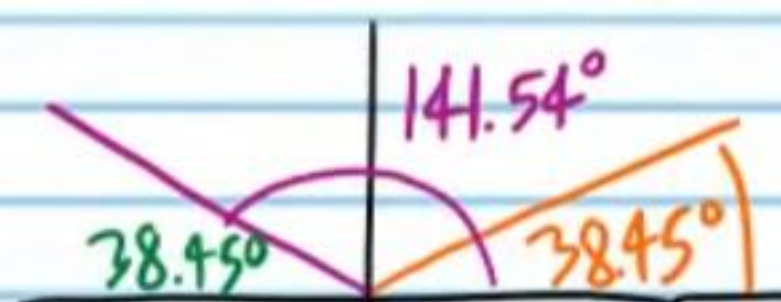
$$\sin B \approx 0.6218...$$

$$B \approx \sin^{-1}(0.6218...)$$

$$B \approx 38.4517^\circ$$

$$38.4517^\circ + 20^\circ < 180^\circ$$

$$58.4517^\circ < 180^\circ \checkmark$$



$$B \approx 141.54829^\circ$$

$$141.54^\circ + 20^\circ < 180^\circ$$

$$161.54^\circ < 180^\circ \checkmark$$

II) Find $m\angle A$.

$$A+B+C=180^\circ$$

$$A + 38.4517^\circ + 20^\circ \approx 180^\circ$$

$$A \approx 121.54829^\circ$$

III) Find a .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 121.54829^\circ}{a} \approx \frac{\sin 20^\circ}{11}$$

$$a \approx \frac{11 \sin 121.54829^\circ}{\sin 20^\circ}$$

$$a \approx 27.408\text{m}$$

$b=20\text{m}, c=11\text{m}, C=20^\circ$

$$B \approx 38.4517^\circ, A \approx 121.54829^\circ$$

$$a \approx 27.408\text{m}$$

II) Find $m\angle A$.

$$A+B+C=180^\circ$$

$$A + 141.54829^\circ + 20^\circ \approx 180^\circ$$

$$A \approx 18.4517^\circ$$

III) Find a .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 18.4517^\circ}{a} \approx \frac{\sin 20^\circ}{11}$$

$$a \approx \frac{11 \sin 18.4517^\circ}{\sin 20^\circ}$$

$$a \approx 10.179\text{m}$$

$b=20\text{m}, c=11\text{m}, C=20^\circ$

$$B \approx 141.54829^\circ, A \approx 18.4517^\circ$$

$$a \approx 10.179\text{m}$$